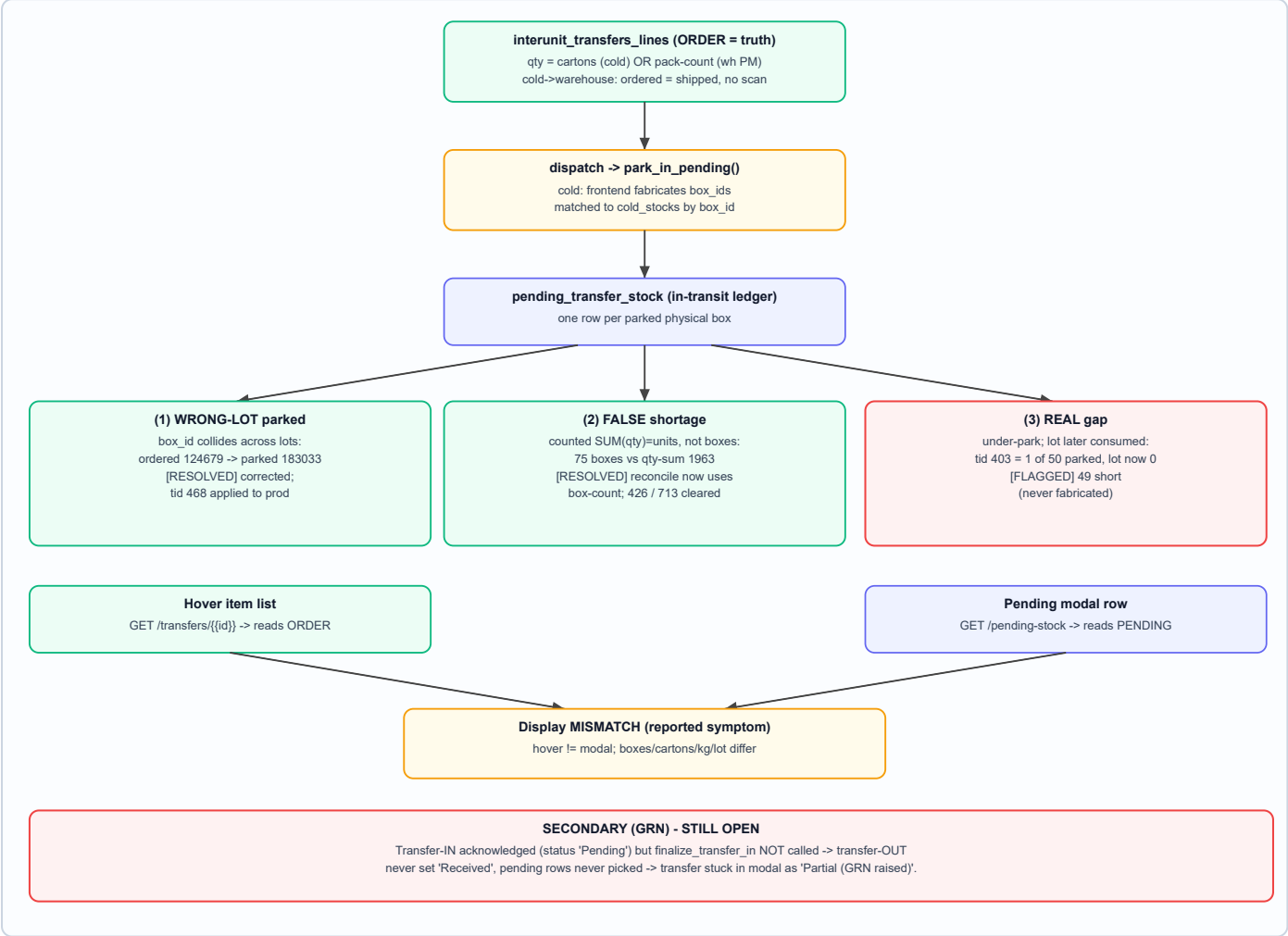


Inter-Unit Transfer — Pending Stock: Root-Cause & System Reconciliation

IMS (ims-app-backend / frontend-) · prod DB warehouse_db · v2 (post-fix), 2026-06-01 · generated from live schema + source. Module prefix: /interunit

1 · Root-Cause Visualization



■ authoritative / correct ■ artifact / gap ■ corruption / mismatch

In one line: One order lives in three tables that were never reconciled, and the in-transit ledger is built from **fabricated frontend box_ids** (cold→warehouse has no real scanning). That produced three mismatch modes — **(1)** wrong-lot parking from box_id collisions, **(2)** false shortages from counting pack-units instead of boxes, and **(3)** genuine under-dispatch gaps — so the hover (reads the order) and the modal (reads pending) disagreed. Modes 1 & 2 are now **resolved**; mode 3 is correctly **flagged**. A separate open item: GRNs acknowledged but not finalized keep transfers stuck as “Partial.”

Item	Status	Detail
Duplicate rows in modal	FIXED (code)	group by dispatch only; show 'mixed' for cross-company
Reconcile keying (item-case)	FIXED (code)	match by lot_no, not item_description
(1) Wrong-lot parking	CLEARED (prod)	tid 468: 50× wrong lot 183033 → 50× ordered lot 124679 (applied 2026-06-01)
(2) False shortage	FIXED (code)	expected = COUNT(transfer_boxes), not SUM(qty); 426 (1888) & 713 (14) cleared
(3) Real gap	FLAGGED	tid 403: 1 of 50 parked, lot 125883 since consumed (handled by tid 778). Stale dispatch to cancel
GRN acknowledged-not-finalized	OPEN	6 transfers stuck 'Partial'; needs auto-finalize on complete acknowledgement

2 · Backend Tables & Constraints (live)

interunit_transfer_requests Transfer request (pre-dispatch ask)

12 rows · 14 cols

Column	Type	Null
id	integer	NO
request_no	character varying(100)	NO
request_date	date	NO
from_site	character varying(100)	NO
to_site	character varying(100)	NO
reason_code	character varying(100)	NO
remarks	text	NO
status	character varying(20)	YES
reject_reason	text	YES
created_by	character varying(255)	YES
created_ts	timestamp without time zone	YES
rejected_ts	timestamp without time zone	YES
updated_at	timestamp without time zone	YES
reason_description	text	YES

Constraints:

- UNIQUE: request_no
- PK: id

interunit_transfers_header Transfer-OUT header (the dispatch / challan)

452 rows · 19 cols

Column	Type	Null
id	integer	NO
challan_no	character varying(100)	NO
stock_trf_date	date	NO
from_site	character varying(100)	NO
to_site	character varying(100)	NO
vehicle_no	character varying(50)	NO
remark	text	NO
reason_code	character varying(100)	NO
status	character varying(20)	YES
request_id	integer	YES
created_by	character varying(255)	YES
created_ts	timestamp without time zone	YES
approved_by	character varying(255)	YES
approved_ts	timestamp without time zone	YES
updated_ts	timestamp without time zone	YES
has_variance	boolean	YES
driver_name	character varying(255)	YES
from_cold_unit	character varying(50)	YES

Column	Type	Null
unallocated_boxes	integer	YES

Constraints:

- UNIQUE: challan_no
- FK: request_id → interunit_transfer_requests.id
- PK: id

interunit_transfers_lines	ORDER lines — authoritative qty per (lot, item)	35457 rows · 19 cols
Column	Type	Null
id	integer	NO
header_id	integer	NO
rm_pm_fg_type	character varying(10)	NO
item_category	character varying(255)	NO
sub_category	character varying(255)	NO
item_desc_raw	character varying(500)	NO
item_id	integer	YES
hsn_code	character varying(50)	YES
pack_size	numeric	NO
unit_pack_size	numeric	YES
qty	integer	YES
uom	character varying(20)	YES
net_weight	numeric	NO
total_weight	numeric	NO
batch_number	character varying(100)	NO
lot_number	character varying(100)	YES
created_at	timestamp without time zone	YES
updated_at	timestamp without time zone	YES
storage_location	character varying(255)	YES

Constraints:

- FK: header_id → interunit_transfers_header.id
- PK: id

interunit_transfer_boxes	Box rows attached to the dispatch (frontend-supplied)	14362 rows · 15 cols
Column	Type	Null
id	integer	NO
transfer_line_id	integer	NO
header_id	integer	NO
box_number	integer	NO
article	character varying(500)	NO
lot_number	character varying(100)	YES
net_weight	numeric	YES
gross_weight	numeric	YES
created_at	timestamp without time zone	YES
updated_at	timestamp without time zone	YES

Column	Type	Null
batch_number	character varying(255)	YES
transaction_no	character varying(255)	YES
box_id	character varying(255)	YES
qr_data	jsonb	YES
storage_location	character varying(255)	YES

Constraints:

- FK: transfer_line_id → interunit_transfers_lines.id
- FK: header_id → interunit_transfers_header.id
- PK: id

pending_transfer_stock IN-TRANSIT ledger — one row per parked physical box 5422 rows · 41 cols

Column	Type	Null
id	bigint	NO
transfer_type	character varying(20)	NO
transfer_out_id	bigint	NO
transfer_out_challan_no	character varying(50)	NO
box_id	character varying(100)	NO
transaction_no	character varying(50)	NO
from_company	character varying(10)	NO
to_company	character varying(10)	NO
from_site	character varying(100)	NO
to_site	character varying(100)	NO
from_storage_type	character varying(20)	NO
to_storage_type	character varying(20)	NO
source_table	character varying(50)	NO
source_row_id	bigint	YES
destination_table	character varying(50)	NO
item_description	character varying(255)	NO
lot_no	character varying(50)	YES
batch_number	character varying(50)	YES
weight_kg	numeric	NO
no_of_cartons	integer	YES
cold_storage_data	jsonb	YES
rm_pm_fg_type	character varying(10)	YES
item_category	character varying(100)	YES
sub_category	character varying(100)	YES
pack_size	numeric	YES
unit_pack_size	numeric	YES
qty	numeric	YES
uom	character varying(20)	YES
net_weight	numeric	YES
gross_weight	numeric	YES

Column	Type	Null
total_weight	numeric	YES
article	character varying(255)	YES
status	character varying(20)	NO
cancel_reason	character varying(255)	YES
cancelled_at	timestamp without time zone	YES
dispatched_at	timestamp without time zone	NO
dispatched_by	character varying(100)	NO
created_at	timestamp without time zone	YES
updated_at	timestamp without time zone	YES
original_box_id	character varying(100)	YES
reconciled	boolean	YES

Constraints:

- UNIQUE: transaction_no
- UNIQUE: box_id
- PK: id

interunit_transfer_in_header GRN / Transfer-IN header (receipt)

409 rows · 14 cols

Column	Type	Null
id	integer	NO
transfer_out_id	integer	NO
transfer_out_no	character varying(100)	NO
grn_number	character varying(100)	NO
grn_date	timestamp without time zone	NO
receiving_warehouse	character varying(100)	NO
received_by	character varying(255)	NO
received_at	timestamp without time zone	NO
box_condition	character varying(50)	YES
condition_remarks	text	YES
status	character varying(50)	NO
created_at	timestamp without time zone	NO
updated_at	timestamp without time zone	NO
inward_transaction_no	character varying(50)	YES

Constraints:

- PK: id
- FK: transfer_out_id → interunit_transfers_header.id
- UNIQUE: grn_number

interunit_transfer_in_boxes Received box rows (per GRN)

27571 rows · 21 cols

Column	Type	Null
id	integer	NO
header_id	integer	NO
transfer_out_box_id	integer	YES
box_id	character varying(100)	NO

Column	Type	Null
article	character varying(255)	YES
batch_number	character varying(255)	YES
lot_number	character varying(255)	YES
transaction_no	character varying(255)	YES
net_weight	numeric	YES
gross_weight	numeric	YES
scanned_at	timestamp without time zone	NO
is_matched	boolean	NO
created_at	timestamp without time zone	NO
updated_at	timestamp without time zone	NO
issue	jsonb	YES
line_index	integer	YES
inward_box_id	character varying(50)	YES
original_box_id	character varying(100)	YES
reconciled	boolean	YES
reconciliation_id	character varying(100)	YES
scan_source	character varying(50)	YES

Constraints:

- FK: header_id → interunit_transfer_in_header.id
- PK: id
- FK: transfer_out_box_id → interunit_transfer_boxes.id

cfpl_cold_stocks Cold storage master (cdpl_cold_stocks is identical)

34921 rows · 28 cols

Column	Type	Null
id	integer	NO
inward_dt	date	YES
unit	character varying(50)	YES
inward_no	character varying(50)	YES
cold_item_mark	character varying(255)	YES
vakkal	character varying(100)	YES
lot_no	character varying(50)	YES
no_of_cartons	numeric	YES
weight_kg	numeric	YES
total_inventory_kgs	numeric	YES
group_name	character varying(255)	YES
item_description	character varying(255)	YES
storage_location	character varying(100)	YES
exporter	character varying(255)	YES
last_purchase_rate	numeric	YES
created_at	timestamp without time zone	YES
updated_at	timestamp without time zone	YES
box_id	character varying(100)	YES

Column	Type	Null
transaction_no	character varying(100)	YES
item_subgroup	character varying(255)	YES
item_mark	text	YES
value	numeric	YES
inward_transaction_no	character varying(100)	YES
auto_created_from_inward	boolean	YES
spl_remarks	text	YES
canonical_warehouse	text	YES
canonical_group	text	YES
canonical_subgroup	text	YES

Constraints:

- PK: id

3 · Endpoint Payload Schemas & Frontend Callers

Method / Path	Handler	Request	Response	Frontend caller
POST /interunit/transfers	create_transfer	TransferCreate { header: TransferHeaderCreate{stock_trf_date, from_warehouse, to_warehouse, vehicle_no, driver_name?, approved_by?, remark?, reason_code?}; lines: TransferLineCreate[] {material_type,item_category,sub_category,item_description,quantity,uom,pack_size,unit_pack_size?,net_weight?,total_weight?,batch_number?,lot_number?}; boxes?: BoxCreate[] {box_number,box_id?,article,lot_number?,transaction_no?,net_weight,gross_weight}; request_id? }	TransferHeaderResponse	interunitApiService.createTransfer / transferform / directtransferform
GET /interunit/transfers/{id}	get_transfer	path: id	TransferWithLines { header..., lines: TransferLineResponse[], boxes: BoxResponse[], grn_records[], from_cold_unit }	PendingTransfersModal hover (fetchLines), transfer/view, transfer/page
PUT /interunit/transfers/{id}	update_transfer	TransferCreate (same as create)	TransferHeaderResponse	interunitApiService.updateTransfer
DELETE /interunit/transfers/{id}	delete_transfer	query: user_email, user_role	TransferDeleteResponse{success,message,transfer_id,challan_no}	PendingTransfersModal cancel
GET /interunit/pending-stock	list_pending_transfers	query: from_site?,to_site?,company?,from_date?,to_date?,search?	{ records: PendingTransferRecord[] {transfer_out_id,transfer_out_challan_no,dispatched_at,from_site,to_site,from_company,to_company,from_storage_type,to_storage_type,total_boxes,total_cartons,total_kg,dispatched_by,status,header_status,unallocated_boxes,updated_ts}; filter_options }	PendingTransfersModal (table), transfer/page summary

Method / Path	Handler	Request	Response	Frontend caller
GET /interunit/pending-stock/by-lot	pending_by_lot	query: lot_no?,item_description?,from_site?,from_company?	{ pending_cartons, pending_kg, box_count, transfers: [{challan_no, cartons, weight_kg, box_count, updated_ts,...}] }	directtransferform 'Pending Transfers' hover / '+N in transit'
GET /interunit/pending-stock/in-transit-by-lot	in_transit_by_lot	query: company?	{ <lot_no>: {cartons, kg, box_count} } (batched map)	cold-storage/dashboard '+N in transit' overlay (NEW)
POST /interunit/pending-stock/backfill	backfill_pending_from_existing_transfers	query: user_email, user_role, dry_run	{ transfers_scanned, transfers_with_existing_pending, boxes_topped_up_by_lot, boxes_unallocatable, reconciled[] }	PendingTransfersModal 'Sync existing' (auto on open)
POST /interunit/transfer-in	create_transfer_in	TransferInCreate{transfer_out_id, boxes[], cold_storage_items?}	TransferInDetail	interunitApiService.createTransferIn
POST /interunit/transfer-in/pending	create_pending_transfer_in	PendingTransferInCreate{transfer_out_id,...}	GRN header (status=Pending)	interunitApiService.createPendingTransferIn
POST /interunit/transfer-in/{id}/acknowledge[-batch]	acknowledge_pending_box[es_batch]	PendingBoxAcknowledge{box_id, transaction_no,...}	ack result (GRN stays Pending)	transfer-in scan UI
POST /interunit/transfer-in/{id}/finalize	finalize_transfer_in	FinalizeTransferIn{box_condition, condition_remarks, cold_storage_items?}	GRN → Received; pick_from_pending; transfer-OUT → Received	transfer-in finalize button (⚠ often skipped)

4 · Frontend → Endpoint Map

Frontend file	Endpoints called
PendingTransfersModal.tsx	GET /pending-stock, GET /transfers/{id}, POST /pending-stock/backfill, DELETE /transfers/{id}
directtransferform/page.tsx	GET /pending-stock/by-lot, GET /categorial-search, GET /box-lookup*, POST /transfers
cold-storage/dashboard/page.tsx	GET /pending-stock/in-transit-by-lot
transfer/page.tsx	GET /pending-stock, GET /transfers/{id}, GET/DELETE /transfer-in/{id}
transferform/page.tsx	GET /categorial-search, GET /box-lookup*, POST /transfers
lib/interunitApiService.ts	CRUD /transfers, /requests, /transfer-in (+pending/acknowledge/finalize/reconciliation)

5 · Fixes Applied

- **Dedupe (code):** list_pending_transfers groups by dispatch only (1 row/challan; 'mixed' when cross-company).
- **Reconcile keyed by lot_no** (not case-mismatched item_description); receiving-aware (skips once a GRN exists).
- **Corrective reconcile, cold (code):** restore wrong-lot/excess rows to source, pull the ORDERED lot FIFO, flag genuine shortage on interunit_transfers_header.unallocated_boxes. No silent substitution.
- **Box-count measure (code):** warehouse "expected" = COUNT(interunit_transfer_boxes), not SUM(qty) — eliminates false shortages where qty is a pack count (e.g. tid 426/713).
- **Applied to prod:** tid 468 wrong-lot corruption corrected (50 boxes re-lotted to the order). Verified: zero wrong-lot/excess corruption remains.
- **New columns:** unallocated_boxes, updated_ts on interunit_transfers_header. New endpoint GET /pending-stock/in-transit-by-lot.
- **Frontend:** shortfall badge + 'Edited' badge (modal), '+N in transit' overlay (dashboard), 'Edited' chip (directtransferform).
- **Tooling:** apply_reconcile.py (confirm-gated, per-tid or --all). 11/11 dependency-free tests pass.
- **OPEN:** (a) GRN auto-finalize for acknowledged-but-not-finalized transfers; (b) cancel stale dispatch tid 403; (c) deploy — all code is uncommitted (resolve the in-repo merge first).